MILITARY SPECIFICATION

CRUTCH, ADJUSTABLE, WOOD (YOUTH AND CHILD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types and two sizes of adjustable wooden crutches (see 6.1).

1.2 Classification. The crutch shall be of the following types and sizes as specified herein:

Type I - Solid wood bows.
Type II - Laminated wood bows.
Size 1 - Youth size.
Size 2 - Child size.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Shipping, Fiberboard.

Military

MIL-W-6110	Wood, Determination of Moisture Content of.

FSC 6515

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center

Drawing 24368	Crutch, Adjustable (youth and child)

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARD

D805-63	Testing Veneer, Plywood, and Other Glued Veneer Constructions.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103, or University Microfilms Inc. Division of Xerox Corporation, 300 North Zeeb Road, Ann Arbor, MI 48106.

3. REQUIREMENTS

3.1 Materials. Materials used in the manufacture of crutches shall meet the requirements specified herein, and as shown on the drawing.

3.1.1 Wood. Wooden parts of the crutches shall be manufactured of solid wood, for type I, as specified in 3.1.1.1; or laminated wood, (bows only) for type II, as specified in 3.1.1.2. Other portions of type II crutches shall be solid wood in accordance with 3.1.1.1.

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3.1.1.1 Type I, Solid wood. Solid wood components shall be straight grain, dense, Northern hardwood. Species shall be commercial white ash, maple, birch, or oak. The average number of annual rings of ring porous woods, if used, shall not exceed 15 per inch of radius. Wood shall be sound, thoroughly seasoned (12 to 18 percent moisture); and shall be free from knots, splits, decay, bird pecks, checks, and any other defects affecting its appearance or which may affect serviceability. Solid wood shall be commercial straight grain, which as used herein shall mean; the grain of the wood shall not cross from one side of the piece to the other side, in less than eight inches of length.

3.1.1.2 Type II, Laminated wood. Laminated wood shall be cut from panels made up from five, seven, or nine plies of veneer as specified herein. This shall be the type of plywood known commercially as parallel grain construction. The resulting bonded panels shall have a dry shear of not less than 1200 pounds per square inch, with a minimum average of not less than 1700 pounds. Wet shear shall not be less than 850 pounds per square inch, with a minimum average of not less than 1200 pounds per square inch. Testing shall be done as specified in 4.4.3. Not less than 10 determinations for each type of shear shall be made from each lot of laminated wood to determine the above averages. There shall be no signs of delamination after soaking in water and prior to testing.

3.1.1.2.1 Face veneers. The face veneers (face and back) shall be hickory, ash, maple, or yellow birch calipering a minimum of 0.039 inch and maximum of 0.050 inch prior to sanding, except for five ply laminated bows each veneer shall be 0.100 $\pm$ 0.003 inch thick, after sanding and prior to gluing. The veneer shall be straight grained and free from any evidence of decay. There shall be no knots, splits, wormholes, bird pecks, or checks evident in the face veneers. The veneer shall have a moisture content of not more than 10 percent at the time of gluing. Moisture content shall be tested as specified in 4.4.2.

3.1.1.2.2 Inner veneers. The inner veneers shall be yellow birch, maple, hickory, or ash. The veneers used shall caliper between 0.060 and 0.130 inch except the five ply laminated veneers shall each be 0.100 $\pm$ 0.003 inch thick, after sanding, and prior to gluing. Final thickness shall be governed by bow and assembly requirements (see 3.1.2 and 3.2.2). The veneers shall not exceed 10 percent moisture at the time of gluing. Moisture content shall be tested as specified in 4.4.2. Inner veneers shall be glued in the same manner as the face veneers,

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3.1.1.2.2 Inner veneers (cont'd)

and in each case, the individual finished sheet of veneer shall be made of components of a single species. When the inner veneers are made up of several pieces of veneer in lieu of a single one piece sheet, the components shall be tapeless spliced, using veneer of one species bonded together on a tapeless splicer. Quality of the inner veneers shall be the same as the face veneers, except small firm knots shall be allowed. There shall be no voids.

3.1.2 Bolts. There shall be three 10-24 commercial stove bolts, 3/16 inch in diameter. Bolts shall be the lengths shown on the drawing. Bolts shall have a head measuring 1/2 inch diameter plus zero, minus 1/16 inch, or a washer conforming to these dimensions shall be used under the head. The same size washer shall be used under the wing nut. A self-locking wing nut shall be used. The wing nut shall be Amerace Esna Corporation's ESNA PART NO. WC-04, or equal. Bolts and washers shall be steel, cadmium or zinc plated. The wing nut shall be zinc base, die cast alloy. Metal parts shall be finished smooth and free of burrs and sharp edges. Wing nuts and washers shall provide a proper fit on the bolt. Bolts shall completely engage wing nuts and protrude a minimum of one screw length (one complete turn) beyond the plastic insert when the crutch is assembled and nuts are hand tightened. The bolt shall not protrude beyond the height of the wing nut.

All bolts shall be given a suitable treatment for relief of hydrogen embrittlement after electroplating.

3.2 Construction. Each crutch shall consist of the arm or supporting piece, two pre-formed bows, hand piece, foot piece, and three stove bolts, fitted with washers and wing nuts. The complete crutch shall be capable of adjustment from 36 to 48 inches for size 1 and 30 to 38 inches for size 2, overall length, when measured from the bottom of foot piece to the top central portion of the armpiece midway between the bows. The sum of the allowed tolerances shall not increase or decrease the overall length more than 1/8 inch. The assembled crutches and component parts shall be as shown in DPSC Drawing 24368.

3.2.1 Armpiece. The solid wood armpiece shall be dimensioned as shown on the drawing for each size crutch. The upper surface shall form the concave arc of a circle having a radius of approximately 8 inches. The lower surface shall have two mortises, spaced equidistant from sides and ends, to receive the tenons of the upper ends of the bows. The edges and corners shall be rounded. As an alternate, the bottom side of the armpiece shall have a concave curve between the mortises. The armpiece must pass the strength test specified in 4.4.1.

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3.2.2 Bows. The bows shall be fabricated from wood as specified in 3.1.1.1 or 3.1.1.2. If supplied with wood as specified in 3.1.1.2 each face veneer shall be a single one piece veneer. The bows shall be the dimensions specified with a square or round tenon, not less than 1/2 inch in length on the upper end. The assembled crutch shall have the wider surfaces of the bows in opposition. Each bow shall have four holes as shown on the drawing, drilled through the center of the wider surfaces, for attaching the handpiece. Two holes shall be drilled in a similar manner at a specified point from the lower end, for attachment of the footpiece. Holes shall be proper clearance size, see drawing, for 3/16 inch stove bolts. The holes shall be accurately drilled for proper alignment when handpiece and foot piece are assembled to bows. Bows of both types shall be pre-formed before assembly.

3.2.3 Handpiece. The handpiece shall be solid wood, cylindrical in shape, tapering toward each end. There shall be a hole drilled through the center from end to end which shall be proper clearance size for the securing bolt. Handpiece shall pass strength test specified in 4.4.1.

3.2.4 Footpiece. The footpiece shall be solid wood, dimensioned as shown on the drawing. Holes shall be drilled, at one-inch intervals, on centers, (18 holes for youth crutch and 12 holes for child's crutch). The holes shall be proper clearance size, see drawing, for 3/16 inch stove bolts. The holes shall be drilled accurately for proper alignment and easy assembly and adjustment of the footpiece and bows. The lower end of the footpiece shall be reduced in size, tapering to a 3/4 inch diameter at the end.

3.3 Assembly. The tenon of each bow shall be firmly anchored with small nails, pegs, or screws into the mortise in the arm or supporting piece. The handpiece shall be secured in position between the side bows with stove bolt, washers, and a wing nut. The lower ends of the two sides bows shall converge with the footpiece between them and be secured with stove bolts, washers, and wing nuts. Each crutch shall be assembled prior to placing in unit package, so that the overall length of the assembled crutch shall be the minimum.

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3.4 Finish. Each wooden piece of each crutch shall have a protective coat of a good grade, clear, transparent spar varnish, which shall be properly applied before assembly and present a uniform hard, smooth surface and be free from bubbles and tackiness. Care shall be taken to prevent excess varnish from blocking clearance holes for stove bolts. If necessary, holes shall be reamed after finishing. A waterproof sealer coat shall be applied prior to the application of the varnish.

3.5 Strength. Crutches shall not break or otherwise fail when subjected to the tests prescribed in 4.4.1.

3.6 Identification marking. Each crutch shall be permanently marked on the inside top of the bow with the name of the manufacturer, contract number, and the lot number. Marking shall be by stamping or stenciling prior to the application of the varnish coat. Stencil ink or paint used shall be specific for the purpose. Marking shall be legible non-fading and durable. Color shall be black. As an alternate, the identification marking shall be placed on the side of the armpiece.

3.7 Workmanship. The workmanship shall be first class throughout. Crutches shall be free from any defects which may impair their serviceability, appearance, or safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.2 Records. Records of examinations, and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, inspection level S-2 for an AQL specified in 4.3.1.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, as shown in table I.

Table I. Sampling for tests

Component	Characteristic	Reqmt.	Test procedure	Unit of product	Insp. level	AQL (Percent defective
End item	Strength	3.5	4.4.1	One individual crutch	S-2*	**
Wood	Moisture content	3.1.1.1	4.4.2	***	--	***
	Shear strength (dry and wet)	3.1.1.2	4.4.3	***	--	***

* A minimum of three units shall be tested, regardless of lot size.
** Acceptance number shall be zero for all sample sizes.
***Each lot of wood shall be tested as specified in applicable test method. Failure to comply shall be cause for rejection of the lot represented.

4.2.3 Samples. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The crutches shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with, but not restricted to, the following classification of defects. Additional defects shall be classified in accordance with MIL-STD-105.

Table II. Categories and defects

Major A		AQL 1.0
101	Wood components not free from warping, splits, splinters, decay, or checks.	
102	Surfaces of wood members not smoothly finished.	
103	Edges or corners of wood members not smoothly rounded.	
104	Armrest joints not securely fitted and properly assembled.	
105	Bolts, nuts, or washers on crutch fail to fit or assemble properly.	
106	Bolts, nuts, or washers not properly protected against corrosion.	
107	Nails protruding on armpiece.	
Major B		AQL 1.5
151	Handle and hardware fail to fit properly into bows in each adjustable position.	
152	Footpiece and hardware fail to fit properly into bows in each adjustable position.	
Minor A		AQL 2.5
201	Varnish finish omitted or not uniform.	
202	Varnish finish not free from tackiness or bubbles.	
203	Footpiece stuck to bow (with varnish).	
Minor B		AQL 4.0
231	Any dimensions not within tolerances specified.	

A. Possible defects listed indicates minimum inspection requirements.
B. Unless otherwise specified, AQL indicates percent defective.
C. Unit of product for inspection purposes shall be the individual crutch, unless otherwise indicated.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Strength. Each sample crutch shall be tested as follows: Secure the crutch in a vertical position and apply a force of 300 pounds, in a vertically downward direction, to the center of the top surface of the armpiece. Remove this force and apply a vertical downward force of 150 pounds to the center of the handpiece. The crutch shall be secured or clamped for tests at any point or points not more than four inches below the top or above the bottom in such a manner that the bows will be completely unstrained.

4.4.2 Moisture content. The moisture content of the wood used in the fabrication of the crutches shall be determined as described in MIL-W-6110.

4.4.3 Shear strength (dry and wet). Test for dry shear strength shall be conducted in accordance with ASTM Designation D805-63, sections 31 through 35. Test for wet shear strength shall be conducted in accordance with ASTM Designation D805-63, sections 31 through 35, except that in this procedure, the test specimens shall be soaked for four hours in water at room temperature and then shall be tested immediately thereafter, while wet.

4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table III. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified, (see 6.2).

5.1.1 Unit of issue. One pair of crutches, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. A sheet of kraft paper shall be interposed between the adjacent faces of each pair of crutches. Each pair of crutches (one unit) shall be packaged in a heat sealed polyethylene bag, 2.5 mils thick, and further packaged in a box of an appropriate size and design, constructed in accordance with PPP-B-636, type CF, class domestic and fabricated of not less than 125 pound test fiberboard. Closure shall be adequate to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Twelve unit packages (12 units) shall be packed in an exterior container designed for type 2 load and constructed in accordance with PPP-B-636, type CF, class weather-resistant, variety DW. Closure, strapping and waterproofing shall be as specified in the appendix of PPP-B-636.

5.2.3 Level B.

5.2.3.1 Exterior container. Twelve unit packages (12 units) shall be packed in an exterior container designed for type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic, variety DW. Closure shall conform to method I or II as specified in PPP-B-636.

5.2.4 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The crutch is intended for use as an aid in ambulation. It is designed for short term usage of hospital patients and convalescents as well as permenently disabled persons.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and size of crutch required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 The crutches by this specification are listed in the Federal Supply Catalog under the following stock numbers:

National Stock Number	Item Identification
6515-00-148-7205	CRUTCH, ADJUSTABLE, YOUTH, WOOD
6515-00-148-7206	CRUTCH, ADJUSTABLE, CHILD, WOOD

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project Number 6515-1949
CS

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-37233 CRUTCH, ADJUSTABLE, WOOD (YOUTH AND CHILD)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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